

JIRAWAT SAIPHET, PH.D.

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TÜBINGEN, GERMANY

Quantum physicist and postdoc researcher.

SKILLS

Expertise: Quantum Computing, Quantum Information, Tensor Networks, Many-body Physics, Machine Learning

Software: TenPy, NumPy, MLX, PyTorch, CCXT, Docker, Git, Whisper, Ollama

Languages: Python, C++, Rust, Shell

EXPERIENCE

PhD student

Universität Tübingen

06/2021 - 08/2025

Tübingen, Germany

- Collaborated with theory and experimental groups in Ulm.
- Developed tensor network simulations of strongly interacting nitrogen-vacancy (NV) centers in diamond.
- Explored the limitations in simulating the dynamics of NV clusters under interaction, disorder, and dissipation.
- Investigated quantum fisher information and entanglement generation, and achieved highly sensitive probe states composed of collective spins using optimal control methods.

Quantum algorithm researcher

Quantum Technology Foundation (Thailand)

02/2021 - 05/2021

Bangkok, Thailand

- Collaborated with Kasikorn Business Technology Group (KBTG).
- Explored an application of quantum algorithms in financial sector challenges, with focus in a loan-loss provision modeled as quadratic constrained binary optimization (QCBO).
- Proposed a quantum-classical hybrid approach.
- Developed a proof-of-concept code to demonstrate the algorithm's functionality on a mocked dataset.

Master student

Mahidol University.

08/2019 - 05/2021

Bangkok, Thailand

- Conducted research on Quantum Approximate Optimization Algorithm (QAOA) using IBM's *Qiskit*.
- Evaluated the QAOA's potential in solving clustering problems.
- Benchmarked QAOA's performance against the classical *K-means* clustering algorithm.

Undergraduate student

Mahidol University.

08/2015 - 06/2019

Bangkok, Thailand

- Conducted research on decoherence suppression in a single qubit by quantum delayed feedback based on the no-knowledge measurement.
- Analytically derived compatible results between the Itô and Stratonovich stochastic master equations.
- Provided analytical solutions for delayed feedback where the measurement operator commutes with a unitary operator, supplemented by numerical results for the general case.

PUBLICATIONS

- [1] J. Saiphet and D. Braun, "Simulation of the dissipative dynamics of strongly interacting nitrogen-vacancy centers with tensor networks," *Phys. Rev. A*, vol. 111, no. 2, p. 22604, Feb. 2025, doi: [10.1103/PhysRevA.111.022604](https://doi.org/10.1103/PhysRevA.111.022604).
- [2] J. Tangpanitanon *et al.*, "Hybrid Quantum-Classical Algorithms for Loan-Collection Optimization with Loan-Loss Provisions," *Phys. Rev. Appl.*, vol. 19, no. 6, p. 64001, Jun. 2023, doi: [10.1103/PhysRevApplied.19.064001](https://doi.org/10.1103/PhysRevApplied.19.064001).

- [3] J. Saiphet, S. Suwanna, T. Chotibut, and A. Chantasri, “Quantum approximate optimization and k-means algorithms for data clustering,” *Journal of Physics: Conference Series*, vol. 1719, no. 1, p. 12100, Jan. 2021, doi: [10.1088/1742-6596/1719/1/012100](https://doi.org/10.1088/1742-6596/1719/1/012100).
- [4] J. Saiphet, S. Suwanna, A. R. R. Carvalho, and A. Chantasri, “Time-delayed quantum feedback and incomplete decoherence suppression with a no-knowledge measurement,” *Phys. Rev. A*, vol. 103, no. 2, p. 22208, Feb. 2021, doi: [10.1103/PhysRevA.103.022208](https://doi.org/10.1103/PhysRevA.103.022208).
- [5] J. Saiphet, A. Chantasri, and S. Suwanna, “Effects of time delay in no-knowledge quantum feedback control,” *Journal of Physics: Conference Series*, vol. 1380, no. 1, p. 12113, Nov. 2019, doi: [10.1088/1742-6596/1380/1/012113](https://doi.org/10.1088/1742-6596/1380/1/012113).

EDUCATION

Eberhard Karls Universität Tübingen

2021 - 2025

Ph.D. | *Physics (quantum metrology)*

Tübingen, Germany

Supervisor: Prof. Daniel Braun

Mahidol University

2019 - 2021

M.Sc. | *Physics*

Bangkok, Thailand

Supervisor: Assoc. Prof. Areeya Chantasri

Mahidol University

2015 - 2019

B.Sc. | *Physics*

Bangkok, Thailand

Supervisor: Asst. Prof. Sujin Suwanna

REFERENCES

Prof. Daniel Braun

Ph.D. Supervisor

daniel.braun@uni-tuebingen.de

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B.Sc. Major Supervisor

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